1) Present the context Free Grammar in Chomsky Normal Form. | Variant 21 Petcov Nicolai FAF-233

1. Original Grammar

$G = (V_N, V_T, P, S)$ $V_N = \{S, A, B\}$

Start symbol - S $V_T = \{a, b\}$

$P = \{$ $S \to aB \mid aAB$
$A \to a \mid B \mid ABaAB$
$B \to aA \mid bS \mid \varepsilon$ $\}$

2. Eliminating ε - productions

$P = \{$ $S \to a \mid aB \mid aA \mid aAB$
$A \to a \mid B \mid BaAB \mid AaAB \mid aAB \mid ABaB \mid BaB \mid AaB \mid aB \mid ABaA \mid BaA \mid AaA \mid aA \mid ABa \mid Ba \mid Aa \mid ABaAB$
$B \to a \mid aA \mid bS$ $\}$

3. Eliminating unit productions

$P = \{$ $S \to a \mid aB \mid aA \mid aAB$
$A \to a \mid BaAB \mid AaAB \mid aAB \mid ABaB \mid BaB \mid AaB \mid aB \mid ABaA \mid BaA \mid AaA \mid aA \mid ABa \mid Ba \mid Aa \mid ABaAB \mid bS$
$B \to a \mid aA \mid bS$ $\}$

4. Eliminating inaccessible and non-prod. symbols

No changes

5. Final CNF Grammar

$G = (V_n, V_T, P, S)$

$V_n = \{S, A, B, T\_a, T\_b, N_0, N_1, \dots, N_5\}$

$V_T = \{a, b\}$ Start symb $= \{S\}$

$P = \{\ S \to T\_a \mid T\_aB \mid T\_aA \mid N_0B$

$A \to T\_a \mid N_1B \mid N_2B \mid T\_aB \mid N_3A \mid$
$T\_aA \mid BT\_a \mid AT\_a \mid T\_b\,S$

$B \to T\_a \mid T\_aA \mid T\_b\,S\ \}$

$T\_a \to a$

$T\_b \to b$

$N_0 \to T\_a A$

$N_1 \to B T\_a$

$N_2 \to N_1 A$

$N_3 \to AB$

$N_4 \to N_3 T\_a$

$N_5 \to A T\_a\ \}$

2) $G = (V_n, V_T, P, S)$ $V_n = \{S, A, B, C\}$

$V_T = \{a, b\}$

$P = \{$
$S \to bA$
$A \to BC$
$B \to b \mid A$
$C \to B \mid a$ $\}$

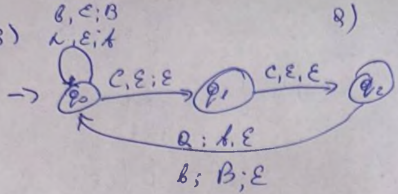
$P' = \{$ $S \to bA$
$A \to BC$
$B \to b \mid BC$
$C \to a \mid b \mid BC$ $\}$

$P'' = \{$ $S \to bA$
$A \to BC$
$B \to bB'$
$B' \to CB' \mid \varepsilon$
$C \to a \mid b \mid bB'C$

$P''' = \{$ $S \to bA$
$A \to bB'C$
$B \to bB'$
$B' \to aB' \mid bB' \mid bB'CB' \mid \varepsilon$
$C \to a \mid b \mid bB'C$ $\}$

$P^{IV} = \{$ $S \to bA$
$A \to bC \mid bB'C$
$B \to b \mid bB'$
$B' \to a \mid b \mid bCB' \mid aB' \mid bB' \mid bB'CB'$
$C \to a \mid b \mid bC \mid bB'C$ $\}$

3) a)



b)

$Q = \{q_0, q_1, q_2\}$ Z_0 = initial stack symb. (ε)

$\Sigma = \{a, b, c, d\}$

$\Gamma = \{A, B, Z_0\}$

$F = \{q_2\}$

1) $\delta(q_0, a, \varepsilon) = (q_0, A)$
2) $\delta(q_0, b, \varepsilon) = (q_0, B)$
3) $\delta(q_0, c, \varepsilon) = (q_1, \varepsilon)$
4) $\delta(q_1, c, \varepsilon) = (q_2, \varepsilon)$
5) $\delta(q_2, a, A) = (q_0, \varepsilon)$
6) $\delta(q_2, b, B) = (q_0, \varepsilon)$

c) abbccbba

Path: $q_0 \to q_0 \to q_0 \to q_0 \to q_1 \to q_2 \to q_0 \to q_0 \to q_0$

Ends in non-final state

• Since this word ends in non-final state q_0, we can't accept this word, because it doesn't end with "cc"

4) $G = (V_N, V_T, P, S)$ $V_N = \{S, A, B, C, D\}$

$V_T = \{a, b, c, d, e\}$

$P = \{ S \to C$ $D \to Be$

$C \to BcA$ $B \to a \mid Bba$

$A \to b \mid dD$ $\}$

sets	FIRST	LAST
S	a	b, e
A	b, d	b, e
D	a	e
B	a	a
C	a	b, e

$a < FIRST(S)$

$LAST(B) = a \Rightarrow a \in LAST(B) \Rightarrow$

$\Rightarrow a > b$

$C \to BcA$

$B = c$

$c < A \Rightarrow c < \{b, d\} \Rightarrow$

$\Rightarrow c < b,\ c < d$

$A \to dD$

$d = D$

$D \to Be$

$B = e$

$B \to Bba$

$B = b$

$b < a$

$B > b$ $b > \$$

$a > \$$ $e > \$$

$\$ < a$

$\$ < a < b < a < c < d < a < e > \$$

$\$ < a < b < a = L = c < d < e > \$$

$\$ < a < b < a = L = c < d = A > \$$

$\$ < a < b < a = L = c = D > \$$

$\$ < a < b < a = L = C > \$$

$\$ < a < b < a = B > \$$

$\$ < a < b < a = B > \$$

$\$ < a < b = B > \$$

$\$ < a = B > \$$

$\$ < C > \$$ $\$ < S > \$$

Matrix

	a	b	c	d	e	S	A	B	C	D	$
a								<			>
b	<										>
c		<		<			<				
d								<			=
e											>
S											>
A											>
B		=	=		=						>
C											>
D											>
$		e								<	

Analysis of word abaedae